Alex Rivera

Principal Firmware Engineer | Motor Control, Simulation and AI-Assisted Development

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Summary

Principal firmware engineer and former engineering manager with experience in motor control, simulation and AI-assisted development for warehouse robots. Software lead on 3 robot launches.

Skills

- **Firmware:** C++, FreeRTOS, CAN, motor control, PID loops, hardware-in-the-loop testing
- **Simulation and tooling:** Python, drive and battery simulation, comparison against logged runs
- **AI-assisted development:** Claude Code, coding-agent workflows, automated gates, LLM evaluation
- **Leadership:** team management, hiring, launches

Experience

Northbridge Robotics **Jun 2017 – Oct 2026**

Principal Firmware Engineer | Madison, WI Jan 2023 – Oct 2026

- Built an internal coding-agent workflow with Claude Code, reviewed by automated gates, that delivered 180 tasks.
- Measured it in a pilot of 8 engineers over 6 weeks. Median review time fell about 25%.
- Wrote an LLM evaluation harness in Python and ran it against 300 past pull requests.
- Replaced a hand-tuned PID loop with a calibrated model, cutting settling time about 35% on the test rig.
- Built a simulation of drive, battery and wheel slip, compared against 60 logged runs.

Firmware Engineering Manager | Madison, WI Mar 2020 – Jan 2023

- Scope: a team of 7 firmware and test engineers. Hired 4 engineers, and 2 were promoted.
- Software lead on 3 robot launches.
- Built a hardware-in-the-loop bench with 212 automated tests, run on every merge.

Senior Firmware Engineer | Madison, WI Jun 2017 – Mar 2020

- Wrote motor-control firmware in C++ for a fleet of 1,400 warehouse robots.
- Cut motion-controller boot time from 42 s to 9 s.

Lakeshore Instruments **Aug 2012 – Jun 2017**

Embedded Software Engineer | Milwaukee, WI Aug 2012 – Jun 2017

- Wrote embedded software for industrial instruments.

Education

- B.S., Computer Engineering, University of Wisconsin–Madison, 2012

Certifications

- Certified ScrumMaster